

HIMAP Program Technology Profile Assessment — Application Enhancement Specification

You are working on an existing web application called the **HIMAP Program Technology Profile Assessment**.

Do **not rebuild the application from scratch**. First inspect the existing codebase, understand the current architecture, data model, state management, scoring logic, accessibility patterns, styling, and components. Preserve existing working functionality unless a change is required by this specification.

The application is intended primarily for **executives, operations leaders, transformation leaders, technology leaders, healthcare delivery leaders, and account leaders** evaluating the technology maturity of healthcare IT-BPM programs.

The experience must feel like an **executive diagnostic and recommendation tool**, not a survey.

1. EXISTING FUNCTIONALITY TO PRESERVE

The following are already built and must remain functional:

- Email, company, and line-of-business entry
- Instructions with required acknowledgment
- Confidentiality notice
- Anonymized-research notice
- Explicit “Do not upload PHI” notice
- Healthcare industry selection, with architecture that can support future industries
- Three healthcare archetypes:
 - A — Non-Clinical Interaction
 - B — Clinical Interaction
 - C — HIM Production
- All 42 questions across six domains
- Yellow / Green / Blue scoring:
 - Yellow = 1
 - Green = 2
 - Blue = 3
- Archetype-specific scoring anchors

- Optional artifact picker
- Evidence notes
- Weighted maturity result (20-100 once scored)
- Program maturity profile
- Current P1-P5 package calculation
- D4-Q5 privacy/security compliance cap
- D5-Q2 Clinical/HIM governance cap
- CSV summary download
- Prototype admin page for future-industry drafts
- Keyboard-friendly controls
- Visible focus state
- High contrast
- Responsive layout
- Reduced-motion support

Do not regress accessibility.

2. SIX ASSESSMENT DOMAINS

Maintain these six domains and weights:

Domain	Name	Weight
D1	Work Execution & Digitization	10%
D2	Performance & Quality Intelligence	15%
D3	Workforce Planning & Optimization	15%
D4	Data Infrastructure & Integration	25%
D5	Decision Intelligence & AI Enablement	20%
D6	Agent Experience & Interaction Execution	15%

Each domain contains seven questions.

Domain raw score: 7-21.

Normalize each completed domain to a 20-100 maturity scale:

$\text{normalized} = 20 + ((\text{rawScore} - 7) / 14) * 80$

Anchor mapping: all Yellow / raw 7 = 20; all Green / raw 14 = 60; all Blue / raw 21 = 100.

A valid scored domain must never display 0. If a domain is incomplete, display blank or "Not Scored" rather than 0.

Round consistently for display but use precise values internally where appropriate.

Domain maturity bands:

- 0–47 = Baseline
- 48–61 = Enhanced
- 62–75 = Above Average
- 76–89 = High Performing
- 90–100 = Tech-Led

For package decision logic, treat Baseline + Enhanced as an under-developed domain (normalized domain score ≤ 61). This package-level classification is separate from the question-level Yellow / Green / Blue anchors.

Keep all thresholds in a central configuration object rather than scattering constants through the code.

3. PROGRAM SCORE

Calculate the overall Program Technology Profile Score using the six weighted normalized domain scores.

Display:

- Overall score: 20-100 for completed assessments
- Incomplete assessment: blank / Not Scored (never 0)
- Program maturity
- Six domain scores
- Six domain maturity levels
- Strongest domain
- Lowest-scoring domain
- Number of under-developed domains (score ≤ 61)
- Number of Blue / Tech-Led domains (score ≥ 90)

Base package by weighted program score:

- 0–47 → P1 Foundation Build
- 48–61 → P2 Operational Digitization
- 62–75 → P3 Intelligence Activation
- 76–89 → P4 AI & Automation Scale
- 90–100 → P5 Tech-Led Optimization

However, **the weighted score must NOT be the sole basis for the final package.**

4. DOMAIN-AWARE PACKAGE ASSIGNMENT

Implement a clearly separated package decision engine.

The system must calculate:

1. Base package from overall score
2. Domain overrides
3. Healthcare compliance/governance caps
4. Final recommended package
5. Lowest-domain solution focus
6. Further-analysis requirement

The logic must be explainable and visible to the user.

Override priority

Evaluate rules in this order.

Rule 1 — Data Infrastructure Foundation Override

If D4 Data Infrastructure & Integration is Yellow:

D4 normalized score ≤ 61

assign:

P1 — Foundation Build

Reason: Data infrastructure is foundational to all downstream automation, intelligence, and AI.

Rule 2 — Broad Foundation Gap

If **three or more domains are Yellow**:

domain normalized score ≤ 61

assign:

P1 — Foundation Build

Reason: The program has broad foundational gaps that need stabilization before advanced technology investment.

Rule 3 — Work Execution / Workforce Override

If either:

- D1 Work Execution & Digitization is Yellow
- D3 Workforce Planning & Optimization is Yellow

and Rules 1–2 have not already assigned P1:

cap the program at:

P2 — Operational Digitization

Reason: A program should not move into intelligence or scaled AI while core workflow execution or workforce management remains substantially manual.

Rule 4 — Tech-Led Qualification

If:

- at least four domains are Blue, AND
- D4 is Blue, AND
- D5 is Blue

then the program may qualify for:

P5 — Tech-Led Optimization

subject to all healthcare governance caps.

Rule 5 — Decision Intelligence Constraint

If the overall score would otherwise result in P5 but D5 Decision Intelligence & AI Enablement remains materially under-developed, cap at:

P4 — AI & Automation Scale

The exact threshold must be stored in configuration for future adjustment.

5. HEALTHCARE COMPLIANCE GATES

These rules override maturity scoring.

D4-Q5 Privacy and Security Gate

If D4-Q5 is Yellow / score = 1:

The program cannot be recommended above:

P2 — Operational Digitization

Display an alert:

Healthcare Governance Gate

“Privacy, security, and PHI controls require remediation before advanced intelligence or AI solutions should be scaled.”

Do not use alarming or punitive language.

Clinical / HIM AI Governance Gate

For:

- Archetype B — Clinical Interaction
- Archetype C — HIM Production

If D5-Q2 AI governance/validation = Yellow / score = 1:

The program cannot be recommended above:

P2 — Operational Digitization

Explain:

“Clinical or coding AI governance requires further development before advanced AI-enabled solutions should be recommended.”

6. SAME SCORE, DIFFERENT PACKAGE PRINCIPLE

This is a core design principle of the application.

Two companies with identical Program Scores can receive different recommendations because the **distribution of maturity across domains matters**.

Build the logic and UI to demonstrate this.

Example:

Company A

Program Score = 74

D4 = 55

Result:

P1 Foundation Build — Data Infrastructure Focus

because D4 is Yellow.

Company B

Program Score = 74

D4 is healthy

D1 = 58

Result:

P2 Operational Digitization — Work Execution Focus

because D1 is Yellow.

Company C

Program Score = 74

No Yellow domain

D2 is the lowest at 65

Result:

P3 Intelligence Activation — Performance & Quality Focus

The score establishes P3 and the lowest domain determines which P3 solutions should be prioritized.

Do not simply show “P1,” “P2,” etc.

Show a complete recommendation such as:

P3 Intelligence Activation — D2 Performance & Quality Focus

7. THREE SEPARATE HEALTHCARE SOLUTION TRACKS

Do not use one generic healthcare solution package.

After assessment, use the archetype selected by the user to load a separate solution catalog.

Create three catalogs:

1. nonClinicalSolutions
2. clinicalSolutions
3. himSolutions

Each must have P1–P5 packages.

Each solution component must contain at minimum:

```
{  
  id,  
  archetype,  
  package,  
  priority,  
  component,  
  domain,  
  title,  
  description,  
  rationale,  
  expectedOutcome,  
  dependencies?,  
  guardrails?,  
  metrics?  
}
```

The UI can initially display only:

- Priority
- Solution Component
- Domain
- Description

Allow expanding a component to see:

- Rationale
- Dependencies
- Healthcare guardrails
- Suggested outcome metrics

8. NON-CLINICAL SOLUTION PACKAGES

Typical scope:

- Member services
- Patient services

- Scheduling
- Billing
- Provider services
- Eligibility/status inquiries
- Administrative contact-center workflows

P1 — Foundation Build

Prioritize:

- PHI/privacy and access-control baseline
- Centralized digital SOP and knowledge repository
- Unified service performance dashboard
- WFM tool activation
- Agent desktop/system-switching audit
- Digital QA scorecards
- Structured case and escalation tracking
- Data-source and manual-movement mapping
- Basic frontline BI capability

P2 — Operational Digitization

Prioritize:

- ACD/CRM/WFM/BI integration
- Automated operational reporting
- WFM-ACD integration
- Schedule automation
- Unified agent workspace
- QA auto-sampling
- Digital coaching
- Intraday management dashboard
- Embedded knowledge
- Digital case orchestration
- Context screen pop

P3 — Intelligence Activation

Prioritize:

- Single operational source of truth
- Continuous data-quality monitoring

- Frontline self-service analytics
- AI-assisted QA
- Leading-indicator dashboards
- Threshold alerts
- Structured wrap-up
- ACW reduction
- Contextual knowledge surfacing
- AI coaching recommendations
- Real-time assist pilot
- Decision recommendation layer

P4 — AI & Automation Scale

Prioritize:

- Broad/100% eligible AI interaction analytics
- Automated coaching triggers
- Predictive staffing risk
- Real-time AI agent assist
- AI summarization
- Next-best-action
- Automated intraday re-optimization
- AI recommendations for operational leaders
- Agent self-service QA portal
- Automated performance intervention workflows

P5 — Tech-Led Optimization

Prioritize:

- AI model performance monitoring
- Model retraining
- Real-time assist tuning
- Governance/privacy framework evolution
- Autonomous low-risk administrative workflows
- Generative AI incubation
- Advanced agent development portal
- Next-generation platform evaluation
- Benchmarking
- Center of Excellence contribution

9. CLINICAL SOLUTION PACKAGES

Typical scope:

- Nurse triage
- Utilization management
- Case management
- Care management
- Behavioral health
- Telehealth
- MTM
- Other clinician-led workflows

Critical principle:

AI may augment clinical work but must not independently make care-affecting decisions. Clinicians retain decision authority.

P1 — Foundation Build

Prioritize:

- Minimum-necessary clinical PHI access
- Digital clinical protocols/criteria repository
- Clinical QA and safety scorecards
- Clinical caseload/capacity baseline
- Clinician desktop/workflow audit
- Clinical AI governance baseline
- Structured clinical escalation workflow
- Clinical data-source mapping

P2 — Operational Digitization

Prioritize:

- EHR/case/criteria integration
- Digitized clinical case orchestration
- Clinical queue/capacity integration
- Integrated clinician workspace
- Clinical QA auto-sampling
- Digital clinical handoffs

- Protocol/criteria desktop integration
- Turnaround and clinical-risk dashboards
- Patient/case context assembly

P3 — Intelligence Activation

Prioritize:

- Trusted clinical operational data layer
- Clinical data-quality monitoring
- Formal clinical AI governance
- AI-assisted clinical QA
- Predictive turnaround/caseload risk
- Clinical documentation assistance
- Contextual clinical knowledge
- Leading-indicator alerts
- Controlled real-time clinical assist pilot
- Clinician-reviewed decision-support recommendations

P4 — AI & Automation Scale

Prioritize:

- Broad clinical quality analytics
- Real-time clinician assist
- Clinical auto-summarization
- Predictive clinical workload risk
- Automated low-risk administrative tasks
- Clinical next-best-action recommendations
- AI explainability
- Clinical AI auditability
- Automated case prioritization
- Clinician self-service quality portal

All care-affecting outputs remain clinician reviewed.

P5 — Tech-Led Optimization

Prioritize:

- Clinical AI safety monitoring
- Model drift monitoring
- Bias monitoring

- Continuous clinician-assist tuning
 - Clinical privacy/governance evolution
 - Generative clinical knowledge/documentation pilots
 - Autonomous non-care workflow expansion
 - Personalized clinician development
 - Clinical AI validation CoE
 - Next-generation clinical platform evaluation
-

10. HIM SOLUTION PACKAGES

Typical scope:

- Medical coding
- Clinical Documentation Integrity
- ROI
- Transcription/editing
- Chart abstraction
- Data integrity
- Records-related production work

Critical principle:

Governed automation may be used for lower-risk work, but **high-impact coding or release decisions must retain defined human validation where required by policy or governance.**

P1 — Foundation Build

Prioritize:

- HIM PHI access and audit baseline
- Coding/CDI/ROI digital standards repository
- HIM quality and production dashboard
- Workqueue/capacity baseline
- Coder/HIM workstation audit
- Digital coding QA
- Physician query workflow
- EHR/encoder/billing data-flow mapping

P2 — Operational Digitization

Prioritize:

- EHR–encoder/CAC–billing integration
- Automated workqueue orchestration
- Integrated coder workspace
- Chart inventory/aging dashboards
- QA auto-sampling
- Digital coder coaching
- Structured CDI/query integration
- Coding knowledge integration
- Access-log standardization
- Controlled CAC/NLP activation

P3 — Intelligence Activation

Prioritize:

- Single HIM operational source of truth
- HIM data-quality monitoring
- Coding AI governance
- CAC/NLP-assisted coding
- AI-assisted coding QA
- Predictive backlog/TAT risk
- Denial/error leading indicators
- Contextual coding knowledge
- CDI opportunity detection
- AI-driven coder education

P4 — AI & Automation Scale

Prioritize:

- Broad AI coding/record analytics
- Governed low-risk autonomous coding
- AI-assisted coding at scale
- Predictive chart prioritization
- Documentation-completeness detection
- Denial-risk prediction
- Automated abstraction

- AI explainability/audit tooling
- Coder self-service QA
- Closed-loop coding intervention workflows

P5 — Tech-Led Optimization

Prioritize:

- Coding AI monitoring/retraining
 - Autonomous-coding guardrail optimization
 - HIM governance evolution
 - Controlled autonomous-coding expansion
 - Advanced CDI intelligence
 - Predictive denial/performance model refinement
 - Generative HIM knowledge/query drafting
 - Next-generation HIM platform evaluation
 - HIM automation CoE contribution
 - Technology roadmap co-development
-

11. LOWEST-DOMAIN SOLUTION MODIFIER

The package determines the maturity stage.

The **lowest domain determines which components inside the package are recommended first.**

Create a mapping.

D1 lowest

Prioritize:

- SOP digitization
- workflow orchestration
- handoffs
- escalation
- action tracking
- process automation

D2 lowest

Prioritize:

- QA platform
- quality analytics
- digital coaching
- education
- AI-assisted QA
- performance intelligence

D3 lowest

Prioritize:

- WFM/capacity tooling
- forecasting
- scheduling
- adherence
- workqueue/inventory control
- predictive staffing

D4 lowest

Prioritize:

- data-source mapping
- integrations
- APIs/ETL
- data quality
- single source of truth
- PHI/security/access controls

D5 lowest

Prioritize:

- BI/self-service
- leading indicators
- predictive intelligence
- AI governance
- recommendation layer
- AI ROI/value framework

D6 lowest

Prioritize:

- desktop/workstation audit
- workspace consolidation
- context delivery
- embedded knowledge
- summarization
- real-time assist

Within the recommended package, sort matching lowest-domain solutions before other components of the same priority level.

12. PRIORITY LEVELS

Each solution must carry a delivery priority.

Use package-appropriate values:

P1–P4

-  Immediate
-  Near-Term
-  Roadmap

P5

-  Ongoing
-  Innovation
-  Strategic

Do not rely on color alone. Always show the text label for accessibility.

13. ASSESSMENT CONFIDENCE

Add a new **Assessment Confidence** measure.

This is separate from technology maturity.

The application must distinguish:

“What does the assessment suggest?”

from:

“Do we have enough evidence to confidently prescribe a solution?”

Calculate an evidence-coverage measure from the 42 questions.

Evidence may include:

- artifact selected
- evidence note
- other valid evidence metadata already supported by the application

Do not inspect or process actual PHI.

Suggested initial bands must be configurable:

High Confidence

85–100%

Meaning:

Recommendation Ready

Enough supporting evidence exists to move into solution validation/planning.

Moderate Confidence

70–84%

Meaning:

Targeted Validation Recommended

The package is directional, but selected findings should be validated.

Low Confidence

Below 70%

Meaning:

Further Analysis Required

Do not present the solution package as a final prescription.

Keep these thresholds in configuration.

14. FURTHER ANALYSIS DECISION ENGINE

Add an **Assessment Disposition**.

Possible statuses:

1. **Recommendation Ready**
2. **Targeted Further Analysis Recommended**
3. **Deep-Dive Assessment Required**

Trigger further analysis based on several conditions.

Do not rely on a single score.

Trigger A — Low evidence

If evidence confidence is below configured threshold:
recommend further analysis.

Trigger B — Package boundary

If overall score is within approximately 3 points of a package threshold:

47/48

61/62

75/76

89/90

flag:

Package Boundary Validation

Explain that a small scoring difference could change the maturity tier.

Make the tolerance configurable.

Trigger C — Domain outlier

If the lowest domain is materially lower than the remaining domain profile, flag:

Domain Deep Dive Recommended

Use a configurable gap, initially approximately 20 normalized points below the mean/median of the other domains.

Example:

D1 = 82

D2 = 85

D3 = 80
D4 = 48
D5 = 79
D6 = 86

Result:

D4 Data Infrastructure Deep Dive Recommended

Trigger D — Bimodal maturity

If multiple domains are highly mature while one or more remain materially underdeveloped, flag:

Mixed Maturity Profile — Targeted Analysis Recommended

Do not rely solely on the weighted average.

Trigger E — Healthcare governance uncertainty

If:

- D4-Q5 is Yellow
- Clinical D5-Q2 is Yellow
- HIM D5-Q2 is Yellow
- evidence supporting either answer is missing/weak

display:

Governance Deep Dive Required Before Advanced AI Recommendation

Trigger F — Mixed healthcare archetype

The application currently scores one line of business per assessment.

If the user indicates that the line of business contains materially different work types such as:

- member services + nurse triage
- clinical + coding
- non-clinical + HIM

recommend:

Assess each line of business/archetype separately.

Do not blend fundamentally different operating models into one maturity score.

15. FURTHER ANALYSIS RECOMMENDATION BY DOMAIN

If further analysis is triggered, tell the executive exactly what should be investigated.

D1

Workflow & Digitization Deep Dive

Review:

- SOP consistency
- workflow variation
- handoffs
- escalation
- manual task inventory
- automation opportunities

D2

Performance & Quality Intelligence Deep Dive

Review:

- QA platform
- sampling
- coaching
- metric definitions
- quality coverage
- performance analytics

D3

Workforce / Capacity Deep Dive

Review:

- forecasting
- scheduling

- work queues
- intraday control
- workload aging
- capacity model

D4

Data & Integration Deep Dive

Review:

- application architecture
- source systems
- manual data movements
- API/ETL capability
- data quality
- reconciliation
- PHI access
- security controls

D5

Decision Intelligence & AI Readiness Deep Dive

Review:

- analytics maturity
- leading indicators
- AI use cases
- AI governance
- validation
- explainability
- ROI
- human-review boundaries

D6

Worker Experience & Technology Workflow Deep Dive

Use archetype-appropriate language:

Non-Clinical: **Agent Experience Deep Dive**

Clinical: **Clinician Workflow Deep Dive**

HIM: **Coder/HIM Workflow Deep Dive**

Review:

- number of systems
 - navigation
 - re-keying
 - knowledge access
 - documentation effort
 - AI-assist readiness
-

16. EXECUTIVE RESULTS PAGE

Redesign the result experience for a senior leader.

Do not initially show a huge technical table.

The first result screen should answer six questions immediately:

1. Where are we?

Show:

Program Technology Score: 74 / 100

Above Average

2. What package do we need?

Show prominently:

Recommended Package

P3 — Intelligence Activation

Then:

Primary Focus: D2 Performance & Quality Intelligence

3. Why did we receive this recommendation?

Show 2–4 concise explanatory bullets.

Example:

- Overall maturity supports Intelligence Activation.

- No foundational package override was triggered.
- Performance & Quality Intelligence is currently the lowest-scoring domain.
- Healthcare governance requirements have been satisfied.

If an override changes the base package, explicitly state it.

Example:

Base package by score: P3

Adjusted package: P1

Reason: D4 Data Infrastructure remains below the required maturity threshold.

This explainability is mandatory.

4. Where are our biggest technology gaps?

Show the six-domain profile visually.

For every domain display:

- score
- maturity level
- relative ranking

Highlight:

- strongest
- lowest
- compliance-sensitive domain

Do not use color as the only indicator.

5. What should we do?

Show a short **Recommended Technology Roadmap**.

Initially show:

Immediate

Top 3–5 actions

Near-Term

Next 3–5 actions

Roadmap

Longer-term actions

Allow:

View Full Solution Package

This opens the complete package components.

6. How confident are we?

Display:

Assessment Confidence

and:

Assessment Disposition

Example:

Assessment Confidence: 78% — Moderate

Targeted Further Analysis Recommended

Then explain:

Recommended Deep Dive: D4 Data Architecture, Integration & PHI Governance

17. EXECUTIVE SUMMARY

Generate a concise executive narrative from deterministic assessment data.

Do not invent facts.

Example structure:

Executive Summary

“ABC Healthcare Services achieved a Program Technology Profile Score of 74, placing the program in the Above Average maturity range.

Based on the distribution of domain maturity, the recommended engagement is P3 — Intelligence Activation, with Performance & Quality Intelligence as the primary improvement focus.

The program demonstrates relatively strong operational technology foundations, but opportunities remain to increase the use of automated quality analytics, predictive insight, and system-generated recommendations.

Assessment confidence is Moderate because supporting evidence was provided for 78% of assessed items. Targeted validation of the Performance & Quality domain is recommended before the final technology roadmap is confirmed.”

Build this from template logic rather than requiring an LLM.

18. PACKAGE DETAILS VIEW

Create a detailed package view.

Header:

P3 — Intelligence Activation

Subheader:

Clinical Interaction

Show:

[Why this package](#)

Short description.

[Primary Domain Focus](#)

Example:

D4 Data Infrastructure & Integration

[Immediate Priorities](#)

Cards or rows containing:

- component
- domain
- rationale
- description
- suggested outcomes

Near-Term

Same.

Roadmap

Same.

Allow filters:

- All
- D1
- D2
- D3
- D4
- D5
- D6

Default:

show the lowest domain first.

19. DO NOT OVERSTATE THE RECOMMENDATION

Throughout the application, avoid wording that implies that an automated assessment constitutes a final technology prescription.

Prefer:

- Recommended direction
- Indicative package
- Priority opportunity
- Targeted validation
- Recommended deep dive
- Solution hypothesis
- Technology roadmap recommendation

Avoid:

- Guaranteed solution
- Definitive prescription
- Must purchase
- Automatically approved

When evidence is weak, clearly state that deeper assessment is required.

20. LEADER-FRIENDLY ASSESSMENT FLOW

Implement or refine this flow:

Step 1 — Welcome

Explain:

Program Technology Profile Assessment

“A structured assessment of how your program uses technology, automation, data, and intelligence to drive performance and scalability.”

CTA:

Start Assessment

Step 2 — Organization

Collect:

- Email
 - Company
 - Line of Business
 - Industry
-

Step 3 — Healthcare Work Type

Select:

Non-Clinical Interaction

Examples: member services, scheduling, billing, provider services

Clinical Interaction

Examples: nurse triage, utilization management, care/case management, telehealth

HIM Production

Examples: coding, CDI, ROI, abstraction, transcription/editing

Make the selection easy for a business leader to understand.

Step 4 — Data Handling / Confidentiality

Present required acknowledgment.

Strongly reinforce:

Do not upload or enter PHI, patient identifiers, medical records, or other sensitive personal health information.

Step 5 — Assessment

Six-domain wizard.

Show:

Domain 2 of 6

and:

Question 4 of 7

Do not show all 42 questions on one long page.

Allow:

- Save progress locally if currently supported
 - Previous
 - Next
 - Review before submit
-

Step 6 — Review

Show completion by domain.

Example:

D1 ✓
D2 ✓
D3 ✓
D4 ✓
D5 ✓
D6 ✓

Show evidence coverage.

Require confirmation before final scoring.

Step 7 — Results

Show executive results page.

Step 8 — Detailed Recommendation

Show package and archetype-specific solution roadmap.

Step 9 — Further Analysis

Only display prominently if triggered.

Show:

Why further analysis is recommended

and:

What should be analyzed

Do not merely say:

“More analysis needed.”

Give the actual domain and scope.

21. EXECUTIVE COMPARISON CAPABILITY

Design the data structures now so the application can later support:

- assessment history
- reassessment
- before/after comparison
- multiple lines of business
- portfolio view
- benchmark view

Do not build enterprise benchmarking unless data is available.

Never fabricate benchmark values.

If there is no real benchmark dataset, label any future benchmark area:

Benchmark data not yet available.

22. DOWNLOADABLE RESULTS

Keep existing CSV export.

Also prepare the architecture for an executive report containing:

- Company
- LOB
- Archetype
- Date
- Program score
- maturity
- six domain scores
- base package
- domain override
- final package
- lowest domain
- assessment confidence
- assessment disposition
- recommended deep dive
- Immediate actions
- Near-Term actions
- Roadmap actions
- healthcare governance notices

If PDF generation is not already in the project, structure the result data so PDF generation can be added without rewriting the scoring engine.

23. ADMIN / CONFIGURATION ARCHITECTURE

Move business logic into configuration/data structures where reasonable.

The app should be capable of supporting future industries without rewriting the scoring engine.

Separate:

- industry
- archetype
- questions
- scoring anchors
- domain weights
- maturity bands
- package thresholds
- package override rules
- compliance gates
- evidence-confidence thresholds
- solution catalogs
- further-analysis rules

Do not hardcode healthcare-specific descriptions inside generic UI components.

Recommended conceptual structure:

```
/config
  scoring
  domains
  packageRules
  confidenceRules

/industries
  /healthcare
    archetypes
    questions
    complianceRules
    solutions
      nonClinical
      clinical
      him
```

Adapt this to the existing architecture rather than forcing this exact folder structure.

24. EXPLAINABILITY

Every final recommendation must be traceable.

Create a recommendation explanation object such as:

```
{
  basePackage,
  basePackageReason,
  triggeredOverrides: [],
  complianceCaps: [],
  finalPackage,
  lowestDomain,
  primarySolutionFocus,
  confidenceScore,
  confidenceLevel,
  furtherAnalysisTriggers: [],
  assessmentDisposition
}
```

Use this object for both:

- UI
- exports

Never calculate a recommendation separately in multiple components.

There must be a **single source of truth** for recommendation logic.

25. TESTING REQUIREMENTS

Add automated tests for package logic.

At minimum test:

Normalization and score band tests

```
7 raw (all Yellow) -> 20
14 raw (all Green) -> 60
21 raw (all Blue) -> 100
Incomplete domain -> blank / Not Scored
47 -> P1
48 -> P2
61 -> P2
62 -> P3
75 -> P3
76 -> P4
89 -> P4
90 -> P5
```

Same-score/different-domain tests

Test three assessments with overall score approximately 74:

- D4 Yellow → P1

- D1 Yellow but D4 healthy → P2
- no Yellow domains → P3

Compliance tests

- D4-Q5 Yellow must prevent package >P2
- Clinical D5-Q2 Yellow must prevent package >P2
- HIM D5-Q2 Yellow must prevent package >P2
- Non-Clinical must not incorrectly receive Clinical/HIM governance cap

Further-analysis tests

- evidence <70 → Deep-Dive Assessment Required
- evidence 70–84 → Targeted Further Analysis
- evidence ≥85 with no other triggers → Recommendation Ready
- score near threshold → package-boundary validation trigger
- major domain outlier → domain deep-dive trigger

Archetype solution tests

Ensure:

- Non-Clinical loads only Non-Clinical solution catalog
- Clinical loads only Clinical catalog
- HIM loads only HIM catalog

26. ACCESSIBILITY

Preserve and enhance the current accessibility implementation.

Requirements:

- WCAG-conscious semantic structure
- Full keyboard navigation
- Visible focus
- no color-only communication
- labels for icons
- appropriate ARIA where necessary
- responsive layout
- reduced-motion support
- accessible data visualization equivalents
- adequate contrast

Red/Yellow/Green/Blue maturity and priority colors must always have accompanying text.

27. UI / VISUAL DIRECTION

The audience is executive.

Design should be:

- clean
- credible
- restrained
- enterprise
- healthcare appropriate
- data-driven
- easy to scan

Avoid:

- game-like scoring
- excessive animation
- too many colors
- overly technical screens
- crowded dashboards
- large blocks of explanatory text on the first results screen

Use progressive disclosure.

Executive overview first.

Detailed technical findings second.

28. TERMINOLOGY

Use these names consistently:

Assessment Program Technology Profile Assessment

Overall metric Program Technology Profile Score

Maturity levels Baseline

Enhanced

Above Average
High Performing
Tech-Led

Packages P1 — Foundation Build
P2 — Operational Digitization
P3 — Intelligence Activation
P4 — AI & Automation Scale
P5 — Tech-Led Optimization

Archetypes Non-Clinical Interaction
Clinical Interaction
HIM Production

Evidence metric Assessment Confidence

Final assessment status Assessment Disposition

Possible dispositions:

Recommendation Ready
Targeted Further Analysis Recommended
Deep-Dive Assessment Required

29. IMPORTANT BUSINESS PRINCIPLE

The application must reflect this philosophy:

The weighted score tells us the overall maturity of the program.

The domain profile tells us why.

The lowest domains tell us where to intervene.

The archetype tells us which healthcare solutions are appropriate.

Healthcare governance tells us what we are allowed to scale safely.

Evidence confidence tells us whether we know enough to recommend the solution or should conduct further analysis first.

Therefore:

Same score does not necessarily mean same package.

and:

Same package does not necessarily mean same solution roadmap.

30. IMPLEMENTATION APPROACH

Before coding:

1. Inspect the current repository.
2. Identify the existing assessment state model.
3. Identify existing score/package calculation functions.
4. Identify existing healthcare archetype configuration.
5. Identify current export logic.
6. Identify accessibility components and patterns.
7. Propose the minimum architectural changes necessary.

Then implement in logical stages:

Phase 1

Centralize scoring and package decision engine.

Phase 2

Add domain overrides and recommendation explanation object.

Phase 3

Add Assessment Confidence and Further Analysis engine.

Phase 4

Add the three healthcare solution catalogs.

Phase 5

Build executive results page.

Phase 6

Build detailed solution roadmap view.

Phase 7

Update CSV/export data.

Phase 8

Add tests.

Do not replace working functionality unnecessarily.

After implementation:

- run existing tests
- run new tests
- run lint/typecheck/build
- fix regressions
- summarize changed files
- explain any assumptions
- identify any business rules that remain configurable or require stakeholder confirmation

31. EXPECTED RESULTS PAGE OUTPUT

The final results page must feel like an **executive technology advisory output**, not a questionnaire score page.

The information hierarchy should be:

Overall Maturity → Recommended Package → Primary Domain Focus → Why → Recommended Actions → Confidence → Further Analysis

The initial results view should resemble the following structure:

PROGRAM TECHNOLOGY PROFILE ASSESSMENT

Company: ABC Healthcare Services

Line of Business: Member Services

Healthcare Track: Non-Clinical Interaction

PROGRAM TECHNOLOGY PROFILE SCORE

74 / 100

Above Average

RECOMMENDED PACKAGE

P3 — Intelligence Activation

PRIMARY TECHNOLOGY FOCUS

D2 — Performance & Quality Intelligence

ASSESSMENT CONFIDENCE

82% — Moderate

ASSESSMENT DISPOSITION

Targeted Further Analysis Recommended

WHY THIS RECOMMENDATION?

Base package by score:

P3 — Intelligence Activation

Domain override:

None

Healthcare governance cap:

None

Final recommended package:

P3 — Intelligence Activation

Lowest-scoring domain:

D2 — Performance & Quality Intelligence

Recommendation explanation:

"The program's overall technology maturity supports Intelligence Activation. No foundational package override or healthcare governance cap was triggered. Performance & Quality Intelligence is currently the lowest-scoring domain and therefore becomes the primary focus of the P3 solution roadmap."

DOMAIN PROFILE

D1 Work Execution	72	Above Average	
D2 Performance & Quality	65	Above Average	PRIMARY FOCUS
D3 Workforce Planning	78	High Performing	
D4 Data Infrastructure	76	High Performing	
D5 Decision Intelligence	69	Above Average	
D6 Agent Experience	77	High Performing	

Strongest domain:
D3 — Workforce Planning & Optimization

Priority domain:
D2 — Performance & Quality Intelligence

Yellow domains:
0

Blue domains:
0

RECOMMENDED TECHNOLOGY ROADMAP

IMMEDIATE

- 1. AI-Assisted Quality Analytics
Domain: D2
Description: Expand quality visibility using controlled AI-assisted interaction analysis while maintaining human calibration and review.
- 2. Leading Indicator Dashboard
Domain: D5
Description: Introduce forward-looking indicators and threshold alerts for performance, quality and operational risk.
- 3. Performance Coaching Intelligence
Domain: D2
Description: Use QA and performance signals to identify targeted coaching opportunities.

NEAR-TERM

4. Contextual Knowledge Surfacing
5. Predictive Forecasting Enhancement
6. Structured Wrap-Up and ACW Reduction

ROADMAP

7. Real-Time Assist Pilot
8. AI Recommendation Layer

[\[View Full Solution Package\]](#)

ASSESSMENT CONFIDENCE

Evidence Coverage:
82%

Confidence:
Moderate

Interpretation:

"The assessment provides sufficient information for a directional recommendation, but selected findings should be validated before final solution design."

FURTHER ANALYSIS

Status:
Targeted Further Analysis Recommended

Recommended Deep Dive:

Performance & Quality Intelligence Diagnostic

Why:

D2 is the lowest-scoring domain and supporting evidence is below the preferred high-confidence threshold.

Suggested scope:

- QA methodology
- QA coverage
- sampling approach
- calibration
- coaching workflow
- performance-data integration
- interaction analytics capability
- existing AI-enabled QA tools

Purpose:

"Confirm the root causes of the D2 maturity gap and validate the appropriate P3 solution components before implementation planning."

[View Analysis Scope]

EXECUTIVE SUMMARY

"ABC Healthcare Services achieved a Program Technology Profile Score of 74, placing the program in the Above Average maturity range. The recommended engagement is P3 — Intelligence Activation, with Performance & Quality Intelligence identified as the primary technology focus. The program demonstrates relatively mature workforce, data and agent-experience capabilities, while opportunities remain to strengthen quality analytics, leading indicators and AI-supported coaching. Assessment confidence is Moderate at 82%; targeted validation of the Performance & Quality domain is recommended before the final technology roadmap is confirmed."

ACTIONS

[Download Executive Report]

[Download CSV]

[View Full Solution Package]

[Review Assessment]

Results-page behavior

The page must progressively disclose information.

First screen / above the fold

Show only the most important executive information:

- Program Technology Profile Score
- Maturity level
- Recommended package
- Primary domain focus
- Assessment Confidence
- Assessment Disposition

A senior executive should understand the assessment outcome within approximately 10 seconds.

Second level

Show:

- Why this package was selected
- Base package
- Any override
- Any healthcare governance cap
- Final package
- Lowest domain

Third level

Show:

- six-domain profile
- strongest domain
- primary opportunity domain

Fourth level

Show:

- Immediate recommendations
- Near-Term recommendations
- Roadmap recommendations

Display only the highest-priority items initially.

Provide a **View Full Solution Package** control to reveal all recommendations.

Fifth level

Show:

- Assessment Confidence
- Evidence coverage
- Further Analysis recommendation
- Recommended diagnostic/deep dive

Final level

Show:

- Executive Summary
- Downloads
- detailed scoring
- evidence review
- technical assessment detail

Do not require an executive user to inspect question-level scoring unless they deliberately choose to drill down.

32. SAME SCORE / DIFFERENT RECOMMENDATION EXPLAINABILITY

The UI must be able to explain why two programs with the same weighted score can receive different packages.

For example:

Scenario A

Program Score: 74

D4 Data Infrastructure: 55

Base package:

P3 — Intelligence Activation

Override:

D4 Data Infrastructure is Yellow

Final recommendation:

P1 — Foundation Build

Primary focus:

D4 Data Infrastructure & Integration

Explanation:

"Although the weighted program score falls within the P3 range, the Data Infrastructure domain remains below the required maturity threshold. Because reliable data and integration are foundational to advanced automation and AI, the recommended starting point is P1 — Foundation Build."

Scenario B

Program Score: 74

D1 Work Execution: 58

D4 is above the Yellow threshold.

Base package:

P3

Override:

D1 Work Execution remains Yellow

Final recommendation:

P2 — Operational Digitization

Primary focus:

D1 Work Execution & Digitization

Scenario C

Program Score: 74

No Yellow domains.

Lowest domain:

D2 = 65

Final recommendation:

P3 — Intelligence Activation

Primary focus:

D2 Performance & Quality Intelligence

The results engine should generate this explanation dynamically from the actual rules triggered.

33. DO NOT RETURN ONLY A PACKAGE NUMBER

A result such as:

"P3"

is insufficient.

The preferred recommendation format is:

P3 — Intelligence Activation

Primary Focus: D2 Performance & Quality Intelligence

Healthcare Track: Non-Clinical Interaction

Where appropriate, also show:

Targeted Further Analysis Recommended

The application should communicate:

1. What maturity level the program has
2. What package is appropriate
3. Why that package was selected
4. Which domain should be addressed first
5. Which healthcare solution track applies
6. What should be done first
7. Whether the evidence is strong enough to proceed
8. Whether a deeper analysis is recommended

This is the required final user experience.

34. EXECUTIVE-LEVEL USER EXPERIENCE AND VISUAL DESIGN

The application will be used by **company leaders, executives, operations leaders, healthcare leaders, transformation leaders, and technology decision-makers.**

The visual and interaction design must therefore feel like a **premium executive advisory platform**, not an internal survey form or academic questionnaire.

The experience should communicate:

- Credibility
- Professionalism
- Simplicity
- Confidence
- Healthcare appropriateness
- Technology sophistication
- Ease of use

The interface should be visually polished but restrained.

Design Principles

Use an executive-level aesthetic characterized by:

- Clean layouts
- Generous white space
- Strong information hierarchy
- Refined typography
- Consistent spacing
- Professional cards and panels
- Subtle depth and borders
- Restrained use of color
- Clear calls to action
- Easy-to-read charts
- Minimal visual clutter

Avoid:

- Gamified UI
- Cartoonish graphics
- Excessive gradients
- Excessive animation
- Bright or distracting colors
- Dense tables as the default view
- Dashboard overload
- Technical jargon when a business-friendly term is available
- Long paragraphs when concise executive summaries will work

The application should feel appropriate if it were shown to a:

- CEO
- COO

- CIO
 - Chief Transformation Officer
 - Healthcare Operations Executive
 - Client Executive
 - Business Unit President
-

35. NAVIGATION

Navigation must be simple and intuitive.

The user should always understand:

1. Where they are
2. What they need to do next
3. How much of the assessment remains
4. How to go backward
5. How to continue
6. How to review the result

Use a clear assessment flow such as:

**Welcome → Organization → Healthcare Track → Instructions → Assessment → Review
→ Results → Recommendations**

During the assessment show:

Domain 3 of 6

and:

Question 4 of 7

rather than presenting all 42 questions on one page.

Provide persistent but unobtrusive actions for:

- Previous
- Next
- Save/continue if supported
- Exit or return to results where appropriate

Do not make the navigation feel like a complex enterprise system.

The user should be able to complete the assessment without needing instructions on how to use the interface.

36. RESPONSIVE DESIGN — LAPTOP AND MOBILE

The application must be fully usable on:

- Desktop computers
- Laptop computers
- Tablets
- Mobile phones

Use responsive design from the beginning.

Do not build a desktop-only interface and shrink it for mobile afterward.

Laptop / Desktop

On larger screens:

- Use available horizontal space intelligently
- Allow summary cards to display side by side
- Display domain comparison charts comfortably
- Use multi-column layouts where appropriate
- Keep primary content within a readable maximum width
- Avoid overly wide paragraphs and tables

For example, the primary result cards may display:

Score | Recommended Package | Confidence

side by side.

Mobile

On mobile:

Stack executive summary cards vertically.

Example:

Program Score

74 / 100

Above Average

then:

Recommended Package

P3 — Intelligence Activation

then:

Primary Focus

D2 — Performance & Quality Intelligence

then:

Assessment Confidence

82% — Moderate

then:

Disposition

Targeted Further Analysis Recommended

Do not force horizontal scrolling for essential information.

Tables should become:

- stacked cards
- expandable rows
- responsive lists

when necessary.

Domain charts and progress indicators must resize cleanly.

Buttons should be large enough for touch interaction.

Important actions should remain easy to reach using one hand on a phone.

37. MOBILE ASSESSMENT EXPERIENCE

For the 42-question assessment, optimize specifically for mobile completion.

Each question should appear as a focused card or section.

Display:

Question 4 of 7

Question text

Then the three choices:

Yellow — Human-Led

Anchor description

Green — Tech-Supported

Anchor description

Blue — Tech-Led

Anchor description

Make the entire answer card selectable, not just a small radio button.

After selecting an answer, clearly indicate the selected state.

Then allow:

Add Evidence

or:

Add Evidence Note

as optional secondary actions.

Do not overwhelm mobile users by showing lengthy evidence controls unless requested.

38. RESULTS RESPONSIVENESS

The Results page must be optimized separately for executive laptop viewing and mobile viewing.

Laptop Result

Preferred structure:

Score	Recommended Package	Confidence
-------	---------------------	------------

74 / 100	P3 Intelligence Activation	82%
----------	----------------------------	-----

Above Average	D2 Primary Focus	Moderate
---------------	------------------	----------

Followed by:

- Why this recommendation
- Domain profile
- Recommended roadmap
- Further analysis

- Executive summary
- Actions

Mobile Result

Preferred structure:

PROGRAM SCORE

74 / 100

Above Average

RECOMMENDED PACKAGE

P3

Intelligence Activation

PRIMARY FOCUS

D2

Performance & Quality Intelligence

ASSESSMENT CONFIDENCE

82%

Moderate

DISPOSITION

Targeted Further Analysis Recommended

Then reveal additional sections through vertical scrolling and optional expansion.

The most important outcome must remain visible near the top.

39. SEND RESULTS TO MYSELF

At the end of the assessment, provide a clear functionality allowing the user to **send the assessment result to the email address they provided at the beginning of the assessment.**

The primary call to action should be:

Email My Results

or:

Send Results to My Email

The destination email should default to the email address entered by the user at the start of the assessment.

Do not require the user to type the email address again unless they choose to change it.

Before sending, show:

Send this assessment summary to:

name@company.com

Provide:

Send Results

and optionally:

Use a Different Email

If changing the address is permitted, validate the new email address.

40. EMAIL CONTENT

The email must be professionally formatted and suitable for a company executive.

Suggested subject:

Your HIMAP Program Technology Profile Assessment Results

The email should contain a concise executive summary rather than all 42 question responses.

Include:

- Company
- Line of Business
- Healthcare Track
- Assessment date
- Program Technology Profile Score
- Maturity level
- Recommended Package
- Primary Domain Focus
- Assessment Confidence
- Assessment Disposition
- Recommended Further Analysis, if applicable
- Top Immediate Priorities
- Link or attachment to the detailed report if supported

Example:

HIMAP Program Technology Profile Assessment

ABC Healthcare Services

Member Services

Non-Clinical Interaction

Program Technology Profile Score

74 / 100 — Above Average

Recommended Package

P3 — Intelligence Activation

Primary Focus

D2 — Performance & Quality Intelligence

Assessment Confidence

82% — Moderate

Assessment Disposition

Targeted Further Analysis Recommended

Immediate Priorities

- AI-Assisted Quality Analytics
- Leading Indicator Dashboard
- Performance Coaching Intelligence

Then:

View Detailed Results

if a persistent secure results URL is supported.

41. EMAIL PRIVACY AND SECURITY

The email functionality must respect the confidentiality model of the application.

Never email:

- Uploaded artifacts
- PHI
- Patient names
- Medical records
- Sensitive evidence files
- User-entered evidence that could contain restricted information

The application already instructs users not to upload PHI. Preserve this requirement.

Prefer emailing:

- Assessment scores
- Executive summary
- Recommendation
- Solution roadmap
- Further-analysis recommendation

If a downloadable detailed report is attached, it must contain only information approved for the assessment report and must not automatically embed uploaded evidence files.

Display a short reminder near the email action:

Results contain assessment findings only. Uploaded evidence is not included in the email.

42. EMAIL DELIVERY EXPERIENCE

When the user selects **Email My Results**:

1. Show the destination email.
2. Confirm the action.
3. Send the report.
4. Display a clear success state.

Example:

Results Sent

“Your Program Technology Profile Assessment has been sent to [name@company.com](#).”

Provide:

Return to Results

Do not expose technical email-service details to the user.

If sending fails, show:

We couldn't send your results. Your assessment is still available on this page. Please try again.

Provide:

Try Again

Do not lose assessment data when email sending fails.

43. EMAIL IMPLEMENTATION REQUIREMENTS

Implement email sending through a secure server-side service.

Do not expose:

- API keys
- SMTP credentials
- email-service secrets

in client-side code.

Create a clear service abstraction so the email provider can be changed later.

The email action should use the application's existing validated result object rather than recalculating the assessment.

Log only the minimum information necessary for delivery and troubleshooting.

Do not log PHI or uploaded evidence contents.

Add rate limiting or equivalent protection against abuse if the application is publicly accessible.

44. DOWNLOAD AND EMAIL ACTIONS

At the bottom of the Results page, provide a polished executive action area.

Recommended actions:

Email My Results

Download Executive Report

Download CSV

Secondary actions:

View Full Solution Package

Review Assessment

The primary action should normally be:

Email My Results

The interface should make it very easy for an executive to finish the assessment and retain a copy without needing to understand file formats.

45. PLEASING AND LOW-FRICTION USER EXPERIENCE

The application should feel easy and pleasant throughout.

Apply these principles:

Reduce Cognitive Load

Show only what the user needs at each stage.

Progressive Disclosure

Executive information first.

Technical details only when requested.

Clear Language

Prefer:

What should we improve first?

over unnecessarily technical labels.

Helpful Context

Provide short explanations where they add value, but avoid large text blocks.

Meaningful Progress

Users should always know how far they are from completion.

Clear Completion

The end of the experience should feel complete:

Assessment Completed

followed by:

- Result
- Recommendation
- Roadmap
- Further Analysis

- Email My Results

46. VISUAL SYSTEM

Create a consistent design system rather than individually styling pages.

Define reusable tokens/components for:

- Typography
- Heading hierarchy
- Card styling
- Borders
- Radius
- Shadows
- Spacing
- Buttons
- Form controls
- Status indicators
- Charts
- Alerts
- Navigation
- Responsive breakpoints

Use a restrained enterprise palette.

Status colors such as Yellow, Green, and Blue may be used for maturity indicators but must never be the only way meaning is communicated.

Examples:

Yellow — Enhanced

Green — High Performing

Blue — Tech-Led

Always display both text and visual state.

47. EXECUTIVE USABILITY STANDARD

Before considering the feature complete, evaluate the application against the following question:

Can a busy executive with no prior knowledge of the assessment open the application on either a laptop or mobile phone, understand what is being asked, complete the assessment, understand the result, know what action is recommended, and email the result to themselves without needing assistance?

If the answer is no, simplify the experience.

The application should require expertise to interpret the **business findings**, not expertise to operate the application.

48. FINAL END-OF-ASSESSMENT EXPERIENCE

The desired end state is:
ASSESSMENT COMPLETE

ABC Healthcare Services

Program Technology Profile Score
74 / 100
Above Average

Recommended Package
P3 — Intelligence Activation

Primary Focus
D2 — Performance & Quality Intelligence

Assessment Confidence
82% — Moderate

Disposition
Targeted Further Analysis Recommended

YOUR PRIORITY ROADMAP

Immediate

1. AI-Assisted Quality Analytics
2. Leading Indicator Dashboard
3. Performance Coaching Intelligence

Near-Term

4. Contextual Knowledge Surfacing
5. Predictive Forecasting

Roadmap

6. Real-Time Assist Pilot
7. AI Recommendation Layer

FURTHER ANALYSIS

Performance & Quality Intelligence Deep Dive Recommended

KEEP A COPY

[Email My Results]

[Download Executive Report]

[View Full Solution Package]

[Review Assessment]

This end state must work elegantly on both **mobile and laptop/desktop screens**.

The user should leave the application with three things clearly understood:

Where the program is today.

Where it should focus next.

What should happen next.